

Dependency and Scope: On the Translation Between Lemmon and Fitch Proofs

Hans Halvorson

Revised draft with Isabelle/HOL verification

Abstract

Lemmon-style and Fitch-style natural deduction are usually treated as notational variants of one another. They are, but the sense in which they are is more delicate than that description suggests, and the delicacy is instructive. A Lemmon line carries an explicit and exact record of the assumptions it rests on; a Fitch line carries no such record, and its dependencies are bounded, but not determined, by the subproofs enclosing it. We define a canonical map δ from Fitch syntax to Lemmon syntax and prove that every correct Fitch proof is sent to a correct Lemmon proof. The converse implication, as originally formulated, is false: because δ forgets excess scope, it can turn an eigenvariable-invalid Fitch object into a correct Lemmon proof. A separate semantic audit shows that the present Fitch turnstile also admits a proof ending in an undischarged top-level subproof. These machine-found boundary cases are stated explicitly rather than hidden by a round-trip test. Translation in the other direction is not *positional*: there are Lemmon proofs whose lines cannot be carried over one for one, in order, to any Fitch proof. The obstruction is structural. Each notation presents a derivation—which is a tree—as a directed acyclic graph, by writing each line once and citing it thereafter; but where Lemmon permits a line to be cited by any later line whatever, Fitch permits citation only along the paths its nesting of subproofs allows.

We give a construction translating any correct Lemmon proof by way of its unfolding. Fresh-name repair handles the eigenvariable conditions for both universal introduction and existential elimination. In Isabelle/HOL we prove that the repaired construction terminates and yields a correct Fitch proof with the same conclusion. We also answer two economy questions negatively. A five-line Lemmon proof can discharge two different assumptions at the same conclusion line; no permutation, no duplication-free layout, and not even the addition of arbitrary auxiliary target lines can represent those source lines in a well-formed Fitch proof. A further formal counterexample shows that laminar dependency regions do not suffice. The Isabelle definitions generate a Haskell and SML kernel, so the executable checker and the proved equations have a common source. The proof checker supplied in `Halvorson/lemmon-checker-main/src` is authoritative for the rule language of *How Logic Works*. Isabelle therefore contains a separate, exact `hl_formula` and `hl_justification` layer matching its datatypes, including Boolean truth constants, equality as a predicate, modus tollens, excluded middle, propositional consequence, and quantifier negation. The formal mirror also totalises Boolean and biconditional cases at which the handwritten program was partial.

1 Introduction

Anyone who has taught both notations knows that a Lemmon proof and a Fitch proof of the same argument look like transcriptions of one another. In the Lemmon style, each line is annotated with the numbers of the assumptions it depends on, and a rule such as conditional proof discharges an assumption by removing its number from the annotation. In the Fitch style, the same information is carried geometrically: an assumption opens a subproof, drawn as a vertical scope line, and discharging it closes the subproof. The two conventions are so evidently doing the same work that the translation between them is generally left as an exercise, or not mentioned.

It is not quite an exercise. The purpose of this paper is to say exactly what the translation is, where it is easy, where it is not, and what the difficulty reveals about the relationship between the two notations.

The central observation is simple enough to state here. In a Lemmon proof, the dependency set attached to a line is *exact*: it is the set of assumptions the line actually rests on. In a Fitch proof, the corresponding information is the set of assumptions of the enclosing subproofs, and this is merely an *upper bound*. A Fitch line may perfectly well sit inside a subproof whose assumption it never uses. Every asymmetry we discuss below follows from this one.

A second observation, less obvious, does most of the work in Section 4. In a Lemmon proof, the textual position of a line carries no semantic content. Lines may be permuted freely, subject only to the requirement that a line be written after the lines it cites. In a Fitch proof, position is semantic: where a line sits determines what may see it, and a line inside a subproof becomes unavailable once that subproof closes. The translation from Lemmon to Fitch must therefore invent positional information that the source does not contain, and it is possible for the source to place demands that no such invention can satisfy.

A third observation concerns the eigenvariable rules. Universal introduction and existential elimination impose freshness conditions not merely on the formulas immediately cited but on assumptions. Lemmon can state those conditions exactly. Fitch has no record of what a line rests on, only of what encloses it, and must state them about scope instead. Since scope may properly include the dependency set, the two conditions come apart, and a naive translation of a correct Lemmon proof may break a Fitch freshness condition. Section 5 exhibits one instance and gives the verified repair.

The structural arguments were first mechanised with a compact internal rule language. The proof checker in the supplied Haskell `src` tree is the authoritative roster intended for *How Logic Works* [1]. The current Isabelle development now represents that roster directly, alongside the internal layer used by the established translation proofs. Section 7 explains the correspondence and the defects that total formal functions made visible. Section 2 fixes notation. Section 3 treats the easy direction and draws from it the redundancy of the dependency column. Section 4 exhibits the two obstructions to a positional translation in the other direction. Section 5 introduces the graph-theoretic vocabulary needed to say what the obstructions have in common, locates the difference between the notations in how each constrains the sharing of lines, gives the construction that translates by unfolding, and repairs it. Section 6 reports the Isabelle proofs and the negative answers to the two economy questions. Section 7 explains the formalisation and generated code, and Section 8 draws the moral.

2 The two systems

Definition 1. A *Lemmon proof* is a finite sequence of lines. A line is a quadruple $\langle \Gamma, n, \varphi, j \rangle$ where n is the line’s number, φ its formula, j its justification, and Γ a set of line numbers—the *dependency set*. A justification is either A (assumption) or a rule together with the numbers of the lines it cites, all of which must be smaller than n .

The rules fall into two groups. Most of them—modus ponens, conjunction introduction, universal elimination—simply pool the dependency sets of the lines they cite. Four of them discharge: conditional proof and reductio each cite an assumption line together with a line derived from it, and disjunction and existential elimination each cite one or two such pairs. For these, the discharged assumption is removed from the pooled set. Thus in

1	(1)	P	A
2	(2)	Q	A
1,2	(3)	$P \wedge Q$	1,2 $\wedge$ I
1	(4)	$Q \rightarrow (P \wedge Q)$	2,3 CP

line (4) discharges assumption (2), which accordingly leaves the dependency set.

In the implementation a line is a record and a justification an enumeration, one constructor per rule. The constructors of the discharging rules are worth noticing: each names an assumption line together with a line derived from it.

```
data ProofLine = ProofLine
  { lineNumber :: Int
  , formula    :: PredFormula
  , justification :: Justification
  , references  :: Set Int -- the dependency set
  }

data Justification
  = Assumption
  | MP Int Int -- modus ponens, two lines
  | AndIntro Int Int
  | CP Int Int -- assumption line, conclusion line
  | RAA Int Int
  | OrElim Int Int Int Int Int -- disjunction, then two such pairs
  | ExistsElim Int Int Int
  | ForallIntro Int
  -- ... fifteen further rules
```

Definition 2. A *Fitch proof* is a finite sequence of *items*, where an item is either a line—a number, a formula and a justification—or a *subproof*, consisting of an assumption line together with a further sequence of items. The *scope* of a line is the set of assumptions of the subproofs containing it. A line may cite another only if the cited line lies within its scope, that is, only if every subproof containing the cited line also contains the citing line.

The corresponding type is mutually recursive, an item being either a line or a subproof containing further items. A rule that discharges cites a subproof, which is written as the pair

of its first and last line—the SubRef below—and this is exactly the pair that the Lemmon rule already names.

```

type SubRef = (Int, Int)

data FitchItem
  = FLine Int PredFormula FitchRule
  | FSub Subproof

data Subproof = Subproof
  { subAssumeLine :: Int
  , subAssumeForm :: PredFormula
  , subBody :: [FitchItem]
  }

data FitchRule
  = FPremise -- never discharged: outermost level
  | FAssume -- opens a subproof
  | FMP Int Int
  | FCP SubRef -- cites the subproof, not two lines
  | FOrE Int SubRef SubRef
  -- ... and so on

```

The same proof in Fitch:

1	P	Premise
2	Q	Assume
3	$P \wedge Q$	$\wedge$ I 1,2
4	$Q \rightarrow (P \wedge Q)$	CP 2–3

Two conventions deserve comment. First, an assumption never discharged is written in Fitch as a premise at the outermost level, so the choice between “premise” and “assumption” is determined by whether the line is discharged later; nothing in the Lemmon proof marks the difference locally. Second, we take universal introduction to carry a side condition on the name generalised upon, rather than to open a subproof with a flagged constant. Both conventions are found in the literature; the first agrees with the condition already imposed in the Lemmon system, and choosing it lets universal introduction translate one line to one line.

3 From Fitch to Lemmon

This direction is unproblematic, and it is worth seeing why, because the reason is exactly what fails in the other direction.

Definition 3. Let F be a Fitch proof. Define $\delta(F)$, the Lemmon proof induced by F , by flattening F into a sequence of lines in order and assigning to each line n the set $\Gamma(n)$ given by: $\Gamma(n) = \{n\}$ if n is an assumption or premise; $\Gamma(n) = \bigcup_m \Gamma(m)$ for m ranging over the lines cited, if the rule at n discharges nothing; and for a discharging rule, the same union with the discharged assumptions removed.

The whole of δ 's arithmetic is the following function, which computes one line's dependency set given those of the lines before it. The discharging cases subtract; every other rule takes the union of what it cites.

```

depsOf :: Map Int (Set Int) -> FitchRule -> Int -> Set Int
depsOf deps r self =
  case r of
    FPremise -> singleton self
    FAssume -> singleton self
    FCP (a, c) -> delete a (look c)
    FRAA (a, c) -> delete a (look c)
    FOrE d (a1,c1) (a2,c2) ->
      unions [ look d, delete a1 (look c1), delete a2 (look c2) ]
    FExistsE m (a, c) -> look m 'union' delete a (look c)
    _ -> unions (map look (citedLines (toLemmonRule r)))
  where
    look n = findWithDefault empty n deps

```

Theorem 4. δ is a total function on Fitch syntax. If F is a correct Fitch proof, then $\delta(F)$ is a correct Lemmon proof.

Proof. The definition of Γ is a recursion on the position of the line, which terminates because a line may cite only earlier lines. That the resulting annotation satisfies the Lemmon conditions on each rule is checked rule by rule; the discharging rules are the only ones where anything happens, and for those the Fitch scope condition guarantees that the discharged assumption is available to be removed. Isabelle proves the implication as `theorem_4_forward`. $\square$

Remark 5. The converse formerly asserted here is false for the present formal definition of correctness. The six-line Fitch object displayed in the proof of Theorem 23 violates universal introduction because $F(a)$ is in scope at line (4). Its δ -image is nevertheless the correct Lemmon proof displayed immediately above it: recomputation gives line (4) dependency set $\{1\}$, whose assumption does not contain a . Thus correctness of $\delta(F)$ cannot imply correctness of F . This does not impugn the forward translation; it shows that a map which forgets excess scope cannot reflect a scope-sensitive eigenvariable condition.

Proposition 6. For each line n of F , the set $\Gamma(n)$ assigned by δ is contained in the scope of n , and the containment may be proper.

The containment may be proper precisely when a line sits inside a subproof whose assumption it does not use. This has a consequence worth recording.

Corollary 7. δ is not injective. Two Fitch proofs that differ only in the placement of a line within subproofs it does not depend on have the same image.

For Fitch proofs satisfying the intended closed-root convention, the round trip through Lemmon and the checked reverse translation behaves as a normalisation rather than an identity: it may return a proof whose subproofs are no wider than they need to be.

The next observation is elementary but, we think, has not been stated. Because the Lemmon rules determine the dependency set of each line exactly—a system that permitted a line to cite more assumptions than it uses would not satisfy Proposition 6—the dependency column is not independent data.

Proposition 8. *Let L be a Lemmon proof and let L^- be the result of deleting its dependency column. Then L is recoverable from L^- .*

Proof. The recursion in the definition of δ makes no use of the dependency column of the source; it uses only the justifications. Applying it to L^- returns L . $\square$

Remark 9. The pedagogical function of the dependency column is thus not to record information the reader could not otherwise obtain, but to require the student to compute it. This is a defensible reason for a notation to carry redundant data, and it is worth distinguishing from the reason a notation might carry data that is genuinely needed.

4 Two obstructions in the other direction

Definition 10. A translation from Lemmon to Fitch is *positional* if it preserves the sequence of lines: the n th line of the Lemmon proof becomes the n th line of the Fitch proof, with the same formula and the corresponding rule.

Positional translation is what one wants, when it is available. It preserves line numbers, so a student may read the two proofs side by side; and it certifies that the two proofs are the same proof in two notations, rather than two proofs of the same thing.

Theorem 11. *There are correct Lemmon proofs with no positional Fitch image.*

We give two, illustrating two independent obstructions. The implementation distinguishes them, and reports which was met and where:

```
data TranslationError
  = NotNested Int Int [Int] -- line, assumption discharged, boxes open
  | OutOfScope Int Int Int -- line, line cited, box that closed over it
  | PremiseInBox Int Int -- line, the box it was written inside
  -- ...
```

1	(1) P		A
2	(2) Q		A
Example 12 (Discharge order).	1,2	(3) $P \wedge Q$	1,2 $\wedge I$
	2	(4) $P \rightarrow (P \wedge Q)$	1,3 CP
		(5) $Q \rightarrow (P \rightarrow (P \wedge Q))$	2,4 CP

Assumption (1) is made first and discharged first. In a positional Fitch image, the subproof opened by (1) would have to contain the subproof opened by (2), since (1) is opened first and subproofs are properly nested; but a subproof may be closed only when every subproof it contains has been closed, and (4) closes (1) while (2) remains open. Fitch permits discharge only in last-in-first-out order, and Lemmon imposes no such restriction.

1	(1) P		A
2	(2) Q		A
Example 13 (Positional trapping).	1	(3) $P \vee R$	1 $\vee I$
	2	(4) $Q \wedge Q$	2,2 $\wedge I$
		(5) $Q \rightarrow (Q \wedge Q)$	2,4 CP
	1	(6) $(P \vee R) \wedge (Q \rightarrow (Q \wedge Q))$	3,5 $\wedge I$

Here the discharge order is unobjectionable. The difficulty is that line (3) is written between assumption (2) and its discharge at (5), and so in any positional image lies inside the subproof opened by (2). It does not depend on (2)—its dependency set is $\{1\}$ —but that is of no help: when the subproof closes at (5), line (3) passes out of scope, and line (6) cites it. In the Lemmon proof nothing goes out of scope at all. Discharging an assumption changes the dependency set of the new line; it does not affect the availability of any earlier line.

Example 13 is the more instructive of the two, and it isolates the second of our opening observations. The Lemmon proof does not say that line (3) is “inside” anything. It is written where it is written, and that is a fact about the page, not about the derivation. A positional translation is obliged to read the page as though position meant something, and here it does not.

Remark 14. Both examples can be repaired by permuting lines. Swapping (1) and (2) in Example 12 yields a proof in which the discharges nest; moving (3) before (2) in Example 13 lifts it out of the subproof. This local success does not generalise: Theorem 28 below gives a permutation-invariant obstruction.

5 Sharing, and two kinds of graph

Both obstructions concern the linear order of lines, and it is natural to ask what becomes of them when the linear order is set aside. Doing so requires a little graph-theoretic vocabulary, which we give in full, since it is not universally familiar.

5.1 Definitions

Definition 15. A *directed graph* consists of a set of *nodes* together with a set of ordered pairs of nodes, the *edges*. We write $m \rightarrow n$ for an edge from m to n , and call m a *parent* of n . A *path* is a sequence of nodes each joined to the next by an edge; a *cycle* is a path returning to its starting node.

Definition 16. A *directed acyclic graph*, or *DAG*, is a directed graph with no cycles. A *rooted tree* is a DAG with a distinguished node, the *root*, such that every node other than the root has exactly one parent and every node is reachable from the root.

The difference that matters here is the one in the parent count. In a tree, each node hangs from exactly one place. In a DAG, a node may have several parents—and when it does, we say the node is *shared*: it does duty in more than one context at once, without being written down more than once.

Definition 17. Let G be a DAG with root r . Its *unfolding* is the tree whose nodes are the paths from r , with the obvious edges. Unfolding replaces each shared node by one copy per context in which it is used. The unfolding of a tree is itself; the unfolding of a DAG with genuine sharing is strictly larger, and may be exponentially so.

5.2 What each notation presents

The object that both notations record is a *derivation*: a tree whose root is the conclusion, whose internal nodes are applications of rules, and whose children are the premises those rules

require. This is Gentzen’s natural deduction in its original tree form, and neither Lemmon nor Fitch writes it down as a tree. Both write it as a DAG, because both write each line once and then refer back to it, which is precisely what sharing is.

Definition 18. The *citation graph* of a Lemmon proof has the lines as nodes, with an edge $m \rightarrow n$ whenever line n cites line m .

Proposition 19. *The citation graph of a Lemmon proof is a DAG, and every line is available to be cited by every later line.*

Proof. Acyclicity is immediate, since a line may cite only lines with smaller numbers. That every earlier line is available is a matter of the notation containing no device by which a line could cease to be. Discharging an assumption alters the dependency set recorded on a *new* line; it does nothing to any line already written. $\square$

Fitch does have such a device.

Definition 20. The *scope tree* of a Fitch proof has the subproofs as nodes, together with a root representing the whole proof, and an edge from each subproof to those immediately contained in it. A line *belongs* to the innermost subproof containing it. A line m is *visible* to a later line n when the node m belongs to lies on the path from the root to the node n belongs to.

Proposition 21. *The scope tree of a Fitch proof is a rooted tree, and the citation graph of a Fitch proof is a DAG in which every edge $m \rightarrow n$ has m visible to n .*

We can now say exactly where the two notations part company, and it is not where the slogan “Lemmon is a graph, Fitch is a tree” would put it. Both present a derivation as a DAG; both share. The difference is that Fitch constrains its sharing and Lemmon does not. In a Lemmon proof any line may be shared with any later line whatever. In a Fitch proof a line may be shared only along the paths its scope tree permits, and once a subproof closes, everything belonging to it is permanently unavailable.

5.3 The translation, restated

The two obstructions of Section 4 can now be seen for what they are. By Proposition 21, translating a Lemmon proof to Fitch means finding a scope tree under which every citation the source makes is permitted. Both examples are cases where no scope tree does: in Example 12 because the discharges demand incompatible nestings, in Example 13 because a line is cited after the only subproof it could have belonged to has closed.

When no scope tree suffices, one may unfold. Unfolding removes sharing, and a DAG without sharing is a tree; a tree imposes no constraint that could be violated, since nothing is shared to be shared wrongly.

A derivation is represented directly as a tree. Where the Lemmon justification names line numbers, the derivation names subderivations, and a line cited twice becomes two subtrees.

```
data Deriv = Deriv { dForm :: PredFormula, dRule :: DRule }
```

```
data DRule
  = DAssume Int -- discharged by an ancestor
```

```

| DPremise Int -- never discharged
| DMP Deriv Deriv
| DCP Int PredFormula Deriv -- discharges the named assumption
| DOrE Deriv Int PredFormula Deriv Int PredFormula Deriv
-- ...

```

Translation is then two functions and a choice between them. The positional translation is attempted first, because it preserves numbering and structure; unfolding is the fallback, and the result records which was used.

```

data Route = Direct | ViaTree

lemmonToFitch :: Proof -> Either TranslationError (Route, FitchProof)
lemmonToFitch prf =
  case lemmonToFitchDirect prf of
    Right fp -> Right (Direct, fp)
    Left _ -> do d <- toDerivation prf
                pure (ViaTree, derivationToFitch d)

```

Definition 22 (The unfolding translation). Given a correct Lemmon proof, unfold its citation graph into a derivation rooted at the last line. Then traverse the derivation, emitting first the premises, at the outermost level and once each however often they are used, and thereafter, for each node, the subderivations its rule requires followed by the line applying it, opening a subproof at each discharge.

Theorem 23. *The construction of Definition 22 does not always yield a correct Fitch proof.*

Proof. Consider

1	(1) $\forall x G(x)$	A
2	(2) $F(a)$	A
1	(3) $G(a)$	1 $\forall E$
1	(4) $\forall y G(y)$	3 $\forall I$
1,2	(5) $F(a) \wedge \forall y G(y)$	2,4 $\wedge I$
1	(6) $F(a) \rightarrow (F(a) \wedge \forall y G(y))$	2,5 CP

which is correct. Line (4) generalises on a , and $\Gamma(4) = \{1\}$; the sole assumption in that set has formula $\forall x G(x)$, in which a does not occur. The Lemmon side condition is satisfied.

The construction returns

1	$\forall x G(x)$	Premise
2	$F(a)$	Assume
3	$G(a)$	$\forall E$ 1
4	$\forall y G(y)$	$\forall I$ 3
5	$F(a) \wedge \forall y G(y)$	$\wedge I$ 2,4
6	$F(a) \rightarrow (F(a) \wedge \forall y G(y))$	CP 2-5

in which line 4 applies universal introduction on a while $F(a)$ stands as an undischarged assumption in its scope. That is contrary to the side condition, so the proof is not correct. $\square$

It is worth being clear that this is not an artefact of the Fitch system we chose in Section 2. One might think to repair matters by restating Fitch’s side condition against the assumptions a line *depends on* rather than those in scope. But a Fitch line does not record what it depends on; that is precisely the difference between the two notations, and the reason the dependency column of a Lemmon proof, though redundant in the sense of Proposition 8, is not idle. Fitch must state its eigenvariable condition against scope, because scope is the only thing it has. The failure is therefore a genuine one: at this rule, Fitch’s coarser bookkeeping is not merely less informative but licenses strictly less.

Nothing is lost in the way of provability. The sequent above is Fitch-provable by other means, and the two systems remain equivalent in the sense of proving the same sequents. What fails is the structure-preserving translation.

There are two ways to repair it. One is to move the offending line: emit each line at the depth of the innermost assumption in its dependency set, so that scope and dependency coincide, and the two side conditions with them. That proposal is not sufficient: Example 13 has disjoint, hence laminar, dependency regions but still has no Fitch image.

The other is to move nothing and rename instead. It is available because the Lemmon side condition says precisely what is needed to license a substitution.

Lemma 24 (Renaming). *Let D be a derivation of $\Gamma \vdash \psi$, and let b be a name occurring nowhere in D . Then $D[b/a]$ is a derivation of $\Gamma[b/a] \vdash \psi[b/a]$.*

Corollary 25. *If in addition a does not occur in Γ , then $\Gamma[b/a] = \Gamma$, and so $\Gamma \vdash \psi[b/a]$.*

Lemma 24 is an instance of the substitution theorem of [2, Thm. 4.5.14]: for a reconstrual $*$ with admissibility conditions Δ for its function symbols, if $\phi \vdash \psi$ then $\Delta, \phi^* \vdash \psi^*$. Renaming a to b is the reconstrual sending $a = y$ to $b = y$, whose admissibility condition $\exists!y (b = y)$ is a theorem and so discharges, leaving $\phi^* \vdash \psi^*$.

That proof does include the case for universal introduction. It is given, however, for a system in which generalisation is on a free variable, the side condition being that x is not free in Γ ; the step goes through because reconstrual preserves free variables, so that x is not free in Γ^* either.

The system of [1] generalises on a name rather than a variable, and there the step is not free. Substituting one name for another *can* introduce into Γ the very name a universal introduction is required to avoid—which is precisely what the free-variable lemma rules out in the other formulation. The condition that b be fresh is what closes that gap, and since it is the step at issue here we give it.

Universal introduction. Suppose $\Gamma \vdash \forall x \psi(x)$ is derived by $\forall I$ from $\Gamma \vdash \psi(c)$, where by the side condition c occurs neither in Γ nor in $\psi(x)$, and suppose the result holds for the latter.

If $c \neq a$, then $\psi(c)[b/a] = (\psi[b/a])(c)$, and the inductive hypothesis gives $\Gamma[b/a] \vdash (\psi[b/a])(c)$. Universal introduction on c applies: c does not occur in $\Gamma[b/a]$, since it did not occur in Γ and the substitution introduces only b , which differs from c because b occurs nowhere in D while c does; and c does not occur in $\psi(x)[b/a]$ for the same reason. So $\Gamma[b/a] \vdash \forall x (\psi[b/a])(x)$, which is $(\forall x \psi(x))[b/a]$.

If $c = a$, then a occurs neither in Γ nor in $\psi(x)$, so $\Gamma[b/a] = \Gamma$ and $\psi(a)[b/a] = \psi(b)$. The inductive hypothesis gives $\Gamma \vdash \psi(b)$. Universal introduction on b applies: b occurs neither in Γ nor in $\psi(x)$, since it occurs nowhere in D at all. So $\Gamma \vdash \forall x \psi(x)$, and since a does not occur in $\forall x \psi(x)$, this is $(\forall x \psi(x))[b/a]$. $\square$

Existential elimination goes the same way, its witness condition being of the same shape; the remaining rules impose no condition on names and follow the pattern of the propositional cases.

Remark 26. The freshness of b is not a convenience, and the contrast with the free-variable formulation shows why. There, reconstrual preserves free variables, so a universal introduction licensed before the substitution is licensed after it. Here, substitution of one name for another can *add* a name to Γ : if b already occurs in Γ , a universal introduction on b somewhere within D may be licensed before the substitution and not after, since the substitution places b among the assumptions that introduction is required to avoid. It is the second case of the proof above that this breaks.

The repair follows, by Corollary 25. Wherever the construction is about to emit a universal introduction whose eigenvariable occurs in an assumption in scope, rename the subderivation feeding it to a name occurring nowhere in the proof—nowhere, and not merely nowhere in the subderivation, for the reason given in Remark 26. Its open assumptions are untouched, since by the Lemmon side condition a does not occur in them. Its conclusion is untouched, since abstraction removes every occurrence of a . And the new name occurs in no assumption whatever, so *a fortiori* in none in scope. Note that the unfolding shares nothing, so renaming within one subderivation cannot disturb any other.

Applied to the proof of Theorem 23, a single line changes:

1	$\forall x G(x)$	Premise
2	$F(a)$	Assume
3	$G(b)$	$\forall E$ 1
4	$\forall y G(y)$	$\forall I$ 3
5	$F(a) \wedge \forall y G(y)$	$\wedge I$ 2,4
6	$F(a) \rightarrow (F(a) \wedge \forall y G(y))$	CP 2–5

Line 3 instantiates to b rather than a , and line 4 generalises on a name that occurs in no assumption anywhere. Nothing else moves.

Theorem 27 (Totality of the repaired unfolding). *Let the construction of Definition 22 be modified as above, using a globally fresh name whenever an eigenvariable would occur in an assumption in scope. So modified, it yields for every nonempty correct Lemmon proof a well-formed and correct Fitch proof with the same conclusion.*

Two things recommend this repair over the first. It is local within each unfolded subderivation: no additional rearrangement or change of subproof shape is required. And it makes no assumption that a duplication-free image exists. The Isabelle proof packages the four structural lemmas of Section 6 with the verified renaming step.

The construction has a cost, which is the cost of unfolding: a line shared k times is derived k times, and in the worst case the result is exponential in the size of the source. There is also a small point of bookkeeping. If a subproof’s conclusion is a line from outside it—as in deriving $Q \rightarrow P$ from the premise P —then the subproof would contain nothing at all, and Fitch requires a reiteration step to bring the line in. Lemmon has no such rule and needs none, by Proposition 19; and the reiterated line is a tautological consequence of the line it repeats, so nothing is added to the system.

6 Formal answers

6.1 The totality theorem

The four obligations suggested by the construction are now machine-checked. In the Isabelle sources they appear as `unfolding_terminates`, `L2_L3`, `L4_derivation`, and `conjecture_26`:

- (L1) *Unfolding terminates.* A correct, nonempty Lemmon proof is converted to a finite derivation tree. The recursion is well-founded because citations point to smaller line numbers.
- (L2) *Citations precede uses.* The traversal emits every subderivation before the rule application that uses it.
- (L3) *Every citation is in scope.* Premises are emitted at the outermost level, while a discharged assumption heads the subproof containing the uses discharged with it. Isabelle proves the complete resulting Fitch object structurally well formed.
- (L4) *Each rule transfers.* The generated line satisfies the corresponding Fitch rule. The proof includes the subproof endpoints required by conditional proof, reductio, disjunction elimination and existential elimination. It also checks the freshness repair for universal introduction and the analogous witness condition for existential elimination.

The final theorem is stronger than termination: it gives a derivation D , proves the generated Fitch proof correct, and proves that its conclusion is the conclusion of the source Lemmon proof. This is Theorem 27, formerly the totality conjecture.

6.2 Why the eigenvariable rules need attention

Theorem 23 turns on universal introduction, but the same audit must be made for existential elimination: its witness name also carries a freshness condition involving assumptions. The remaining rules place their substantive conditions on the formulas cited and the formula concluded. Arbitrariness, by contrast, is a matter of what a line rests on. Lemmon can say this exactly, because a Lemmon line records its dependencies. Fitch can only say it about scope. Wherever scope properly includes the dependency set—which by Proposition 6 it may—the two conditions can come apart. The formal rule-transfer proof checks both eigenvariable cases.

That the repair is a renaming rather than a rearrangement is a point in favour of the diagnosis. The difficulty is not that the line is in the wrong place; it is that the *name* is doing double duty, standing both as the arbitrary individual of the generalisation and as the individual the enclosing assumption is about. Lemmon can tell those two roles apart because it knows which assumptions the line answers to. Fitch cannot, and so a name that is arbitrary in the one sense is not visibly arbitrary in the other. Giving the generalisation a name of its own removes the ambiguity without touching the structure of the proof at all.

6.3 Negative answers to the economy questions

Theorem 28 (Permutation does not always suffice). *There is a correct Lemmon proof such that no citation-respecting permutation of its lines has a positional Fitch image.*

Theorem 29 (Duplication-free images do not always exist). *There is a correct Lemmon proof with no well-formed Fitch image containing each source line exactly once.*

Both negative answers have the same five-line witness:

1	(1)	P	A
2	(2)	Q	A
1,2	(3)	$P \wedge Q$	1,2 $\wedge$ I
2	(4)	$P \rightarrow (P \wedge Q)$	1,3 CP
1	(5)	$Q \rightarrow (P \wedge Q)$	2,3 CP

Lines (4) and (5) demand subproof references (1,3) and (2,3): two different assumptions are discharged at the same conclusion line. But in a well-formed Fitch proof two distinct subproofs cannot have the same last line. This obstruction is invariant under permutation, proving Theorem 28; and it prevents any arrangement containing each source line once, proving Theorem 29.

The Isabelle result is stronger. Let a target *cover* the source when every source line occurs with its number, formula, and corresponding citation-bearing rule, while arbitrary additional target lines are allowed. Even then the displayed proof has no well-formed Fitch image. Auxiliary lines cannot change the fact that the two source discharge rules require different subproofs ending at line (3). Thus duplication or administrative additions alone cannot repair the obstruction; a more liberal translation must unfold or otherwise rewrite source occurrences.

There is also a negative answer to the proposed laminarity criterion. The dependency regions of Example 13 are disjoint, hence laminar, but the Isabelle theorem `theorem_10_ex12` proves that it has no Fitch image at all. Laminar dependency sets are therefore not sufficient for a duplication-free image.

7 Machine verification

The development is formalised in Isabelle/HOL. Formulas, proof lines, Fitch subproofs, all rule checks, both translations, and the predicates used in the negative results are executable definitions. For example, the paper's word *positional* is represented by an ordinary predicate, not by an informal label:

```
definition positionalImage ::
  "fitch_proof => lemmon_proof => bool" where
  "positionalImage F P =
    list_all2 lineMatches (flatten F) P"
```

Here `lineMatches` requires equality of line number and formula and the corresponding rule; `fitchWF F` is asserted separately whenever the target is required to be an actual Fitch proof. Keeping those two clauses separate proved useful: without well-formedness, any list of Lemmon lines could be laid out flat and called a positional image.

The repaired totality claim is a theorem over arbitrary source proofs, not an extrapolation from examples:

```
theorem conjecture_26:
  assumes "lemmonCorrect P" and "P \<noteq> []"
  shows "\<exists>d. toDerivation P = Inr d
    \<and> fitchCorrect (derivationToFitch d)"
```

```
\<and> fitchConclusion (derivationToFitch d)
    = conclusion P"
```

The retained name `conjecture_26` records its correspondence with the earlier draft; its Isabelle status is `theorem`. The other machine-checked answers are recorded under the following source names:

```
theorem_10
conjecture_27_false
conjecture_28_with_auxiliaries_false
```

The first proves the two original witnesses have no positional image, the second refutes the permutation claim, and the third proves the stronger auxiliary-line result. This is one reason the formal work is helpful philosophically: apparently modest changes in the meaning of *same proof*—same order, same occurrences, or merely coverage of source lines—become distinct mathematical claims.

7.1 The authoritative *How Logic Works* checker

The Haskell source is not treated as disposable prototype code. Its logical datatypes and rule decisions are the specification for the executable *How Logic Works* checker. Isabelle represents them without first translating them into the compact formula and rule types used in the earlier proofs:

```
datatype hl_formula =
  HL_Predicate hl_name "hl_term list"
| HL_Boolean bool
| HL_Not hl_formula
| HL_And hl_formula hl_formula
| HL_Or hl_formula hl_formula
| HL_Implies hl_formula hl_formula
| HL_Iff hl_formula hl_formula
| HL_ForAll hl_name hl_formula
| HL_Exists hl_name hl_formula
```

The corresponding `hl_justification` datatype has exactly the twenty-one Haskell constructors. Line numbers and dependency sets use integers, as Haskell’s `Int` does. The predicate `hlCorrect` reproduces the source checker’s full-proof lookup, uniqueness, forward-number test, formula patterns, and dependency equations. A strengthened `hlVerifiedCorrect` additionally requires positive increasing source order and that every dependency really names an assumption. This distinction is deliberate: exact reproduction makes source-level comparison possible, while the strengthened predicate states the invariant needed for theorems.

The Haskell-shaped Fitch datatype is likewise represented directly. Its total map to Lemmon notation is written with the same delta:

```
abbreviation hlDelta ::
  "hl_fitch_proof => hl_proof" ("\"<delta>\<sup>H _") where
  "\"<delta>\<sup>H F == hlFitchToLemmon F"
```

Reiteration becomes `HL_PropTaut [m]` under this map, just as in `FitchConvert.hs`. Executable Isabelle lemmas check representative modus-ponens and conditional-proof images and one accepted proof for every authoritative justification constructor.

The converse direct route from `FitchConvert.hs` is now represented on the same source-shaped datatypes as `hlLemmonToFitchDirect`. A successful result is proved positional: flattening it preserves every source line number and formula in order. The Isabelle version also incorporates the structural repairs established by the compact development: reversed or multiply reused boxes, late premises, bad subproof scope, and the stronger Fitch eigenvariable-in-scope condition are diagnosed before an image is returned. The executable conditional-proof example verifies both the generated Fitch proof and the round trip under δ_H .

The derivation-tree route from `FitchConvert.hs` is represented on these same datatypes as well. The public operation `hlLemmonToFitchChecked` first accepts a direct image only when the exact Fitch checker verifies it, and otherwise unfolds the citation graph, repairs eigenconstants, and lays out nested subproofs. An executable shared-discharge example forces this fallback and verifies both the resulting Fitch proof and its conclusion after applying δ_H .

That example exposed one further defect in the handwritten fallback. Its single global `boxed` set labels an assumption as discharged if any line of the source proof discharges it, even when that line lies on a dead sibling branch omitted from the derivation of the selected conclusion. The resulting tree can drop a genuine open premise and later cite line zero. Isabelle’s `hlClassifyAssumptions` instead carries the dischargers on the actual ancestor path, which is the condition the tree representation requires.

7.2 Executable tests and the regression corpus

Before the universal results were proved, the translations were implemented in Haskell against the proof checker for *How Logic Works*. Those tests remain useful as concrete validation of the parser, checker, renderer, and examples, but they are no longer being asked to stand in for a proof. The Isabelle mirror now shares their formula and justification structure. The existing application has not yet been mechanically linked against the generated module, so application-level tests and Isabelle-level executable lemmas remain separate regression boundaries.

The current regression source contains 80 cases, of which 29 are marked valid. Twenty-four are expected to admit a positional translation, and for each the round trip is required to return the original proof line for line, *including its dependency sets*—an instance of Proposition 8, since the sets were recomputed and not copied. Five are expected to use the derivation-tree route: the obstruction examples, a premise placed inside a subproof, an ordinary reductio whose premise occurs after the discharged assumption, and the eigenvariable example. For those cases the suite requires the recovered Lemmon proof to check and to have the same conclusion, rather than to preserve line numbers.

Sixteen further proofs, transcribed from photographs of student work, were translated. Fourteen were well formed; all fourteen admitted positional translations and round-tripped exactly. The remaining two contained malformed justifications on the page—a rule cited with the wrong number of lines—and were rejected by the parser, as they should be.

We record the empirical finding for what it is worth: neither obstruction occurred in any student proof. They are not artefacts, as Section 4 shows, but they appear to be rare in practice.

One methodological point deserves recording, because it cost us two false negatives. The

round trip is a weaker test than it appears. Since δ recomputes dependency sets from the rules rather than reading them off the page, a Fitch proof that is *malformed*—one violating the conditions in Definition 2—may still return the Lemmon proof it came from, exactly. Two proofs in the corpus did precisely this, both by placing a premise inside a subproof, where its dependency set recomputes correctly while the subproof appears to derive the premise from its assumption. Neither was detected by the round trip; the first was noticed by looking at the rendered output. The remedy is to check the conditions of the notation directly, and separately:

```
-- Conditions the notation imposes, which a round trip cannot detect:
-- * a premise may occur only at the outermost level;
-- * a line may cite only what is in scope.
fitchWellFormed :: FitchProof -> Maybe String
```

The moral generalises beyond this case. A test that verifies an invariant is preserved under translation cannot, by itself, establish that either side satisfies the conditions of its own notation.

7.3 Semantic audit

The Isabelle development gives both formal layers a standard model-theoretic semantics. For the compact internal language it proves soundness of every derivation-tree rule (`derivOK_sound`) and semantic soundness of the Lemmon turnstile (`lemmon_semantic_soundness`). More importantly for the executable checker, the Haskell-shaped layer now has a complete soundness argument of its own. The local theorem `hlRuleOK_sound` covers every one of the twenty-one authoritative justification constructors. A well-founded induction over cited line numbers then proves `hlVerifiedCorrect_line_sound`, and the resulting end-to-end theorem `hlVerifiedCorrect_sound` says that if all open premises are true, the checked conclusion is true in every standard interpretation.

The induction is stronger than a fixed-model argument. Universal introduction and existential elimination temporarily reinterpret a fresh constant, so the induction hypothesis must apply uniformly to every standard interpretation on the same domain. Making that requirement explicit was one of the useful consequences of formalising the checker rather than merely testing representative proofs.

The Haskell-shaped semantics must also treat the predicate name “=” as identity for equality introduction and elimination to be sound; the handwritten `ModelSemantics.hs` currently treats it as an arbitrary predicate relation. Formalising the authoritative layer makes that hidden model condition explicit. The Isabelle theorem suite records the principal checks under the following names:

```
hlPropositionalConsequence_sound
hl_haskell_model_semantics_needs_equality_condition
hlQuantifierNegationEquivalent_sound
hlEqualityElimination_sound
hlRuleOK_sound
hlVerifiedCorrect_line_sound
hlVerifiedCorrect_sound
```

The first connects the executable `PropTaut` valuation search to arbitrary interpretations. The second puts the accepted equality- introduction proof beside a refuting arbitrary-equality model. The last three lines of the list give local, line-by-line, and complete-proof soundness, respectively; the intervening specialized lemmas discharge quantifier negation and equality replacement inside the local proof.

For executable code Isabelle keeps this mathematical interpretation separate from the finite, error-aware representation

```
HL_FiniteModel
```

corresponding to `ModelSemantics.hs`. Its evaluator returns `None` for an unbound variable, a missing interpretation, or an empty quantifier domain, and it handles biconditionals explicitly. The companion `hlToDNF` operation reproduces the propositional DNF pipeline while repairing the omitted biconditional traversal.

The same audit found a second, independent boundary case at the root of a Fitch proof. Isabelle records it as `fitch_semantic_soundness_fails`. The result should be read constructively: the Fitch checker needs a closed-root condition if its turnstile is to take the premises of the outer level as the assumptions of the whole proof. Formalisation is valuable here precisely because it does not silently supply the intended convention.

7.4 Generated Haskell and SML

The cap theory `LF_All.thy` is also the code-generation entry point. Isabelle exports Haskell and SML from the same executable constants used in the proofs. The generated API now contains two clearly named layers: the compact translation development, and the authoritative `HL_` datatypes, checker, finite evaluator, DNF conversion, δ_H , repaired direct translation, and complete derivation-tree fallback. Generated files are derived artefacts; the Haskell program is authoritative for the intended *How Logic Works* rule choices, while Isabelle is the authoritative, total specification and proof environment for their formal reproduction.

The source audit also found partial cases that a total Isabelle function could not silently inherit. `ProofTypes.hs` omitted `Boolean` from several recursive operations; `ModelSemantics.eval` omitted biconditionals; and `PropDNF` omitted biconditionals from its traversal. Isabelle supplies all of those cases. It also exposes two structural choices in the handwritten checker: LEM does not constrain its dependency set, and the forward-reference test compares numbers without requiring the file itself to be in increasing order. The exact predicate records those behaviors; the verified predicate adds the intended invariants.

Copying the generated module beside the web application still does not replace its parser, JSON codecs, OCR/transcription pipeline, LaTeX rendering, web routes, static pages, or corpus. Those are interface code, not products of Isabelle’s code generator. An application adapter is therefore still needed, but no further rule-roster decision is pending: the supplied Haskell roster is the one to which that adapter must connect.

8 What kind of equivalence is this?

It is tempting to summarise all of this by saying that Lemmon and Fitch notation are inter-translatable, and to leave it there. The results above suggest that the summary conceals more

than it reports.

Translation from correct Fitch proofs to Lemmon proofs is total, canonical, and not injective (Theorem 4, Corollary 7). Translation from Lemmon to Fitch is not positional (Theorem 11), but it is total by way of the repaired unfolding (Theorem 27). Theorems 28 and 29 show that neither permutation nor a duplication-free target can serve in general.

It would be premature to describe the current formal objects as a retract. That formulation requires the converse direction of the original Theorem 4, and Remark 5 gives a counterexample to it. Once the intended closed-root condition is made part of the Fitch turnstile, the appropriate normalisation or retract theorem can be stated and proved against that strengthened notion. What is already proved is the substantive translation result: every correct Lemmon proof has a correct Fitch derivation with the same conclusion, and every correct Fitch proof has a correct Lemmon image.

The reason is that the two notations record different amounts. A Lemmon proof carries its dependency structure exactly, and carries no positional information at all. A Fitch proof carries its dependency structure only up to an upper bound, and carries positional information in addition—information which is, from the point of view of the derivation, arbitrary. Neither is a richer notation than the other. They are differently lossy.

That is a more interesting relationship than notational variance, and it is the sort of relationship that arises whenever two formalisms are said to say the same thing. The case is unusually clean: both formalisms are small, completely specified, and mechanically checkable, so the question of what survives translation can be settled rather than argued. We suggest it is worth having such a case in view when the same question is asked of theories, where it cannot.

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